

ECON 351 - Labor-Leisure Problems

All practice problems + drills to solve

Sources checked (uploaded course materials):

- Homework 1 Ch.4: No labor-leisure exercises
- Homework 2 Ch.9: No labor-leisure (Ch.9 uncertainty only)
- Sample Questions Ch.4: No labor-leisure questions found
- Sample Questions Ch.9: No labor-leisure questions found
- Mock Midterm 1: No labor-leisure questions found
- Econ351 Lecture (Ch.9 upload): Chapter 9 only - no labor-leisure
- Practice Tests 1-3: Q3 & Q4 - Maria problems (included below)

6 problems in this PDF (2 from practice tests + 4 extra drills).

Labor-Leisure Formula Sheet

Time: $T = L + I$ (total hours = labor + leisure; I = leisure hours)

Budget: $c = w \cdot L + V = w \cdot (T - I) + V$

Optimum: $MU_I / MU_c = w$

Cobb-Douglas $U = c^a \cdot I^b$: at optimum $b \cdot c = a \cdot w \cdot I$

Special case $a = b = 0.5$: $c = w \cdot I$ and $I^* = (w \cdot T + V) / (2w)$

Labor hours: $L^* = T - I^*$

Consumption: $c^* = w \cdot I^* + V$

Problem 1: LL-1

Practice Tests 1, 2, 3 - Question 3

Maria has 16 hours per day for leisure (l) and work.

Wage $w = \$20/\text{hr}$, non-labor income $V = \$80$.

Utility $U(c, l) = c^{0.5} * l^{0.5}$, where $c = w * (16 - l) + V$.

- Write the budget constraint.
- Find optimal leisure l^* .
- Find labor hours L^* and consumption c^* .

Show your work below:

Problem 2: LL-2

Practice Tests 1, 2, 3 - Question 4

Same setup as LL-1, but wage rises to $w = \$30/\text{hr.}$

- (a) Find new l^* , L^* , and c^* .
(b) Does Maria work more or less than when $w = \$20$? Explain.

Show your work below:

Problem 3: LL-3

Extra drill - standard Cobb-Douglas

Alex has $T = 8$ hours/day, $w = \$25/\text{hr}$, $V = \$0$.

$$U(c, l) = c^{0.5} * l^{0.5}, \text{ where } c = w*(8 - l).$$

Find l^* , L^* , and c^* .

Show your work below:

Problem 4: LL-4

Extra drill - non-labor income change

Same as LL-3 but $V = \$40$ (lottery winnings).

Find l^* , L^* , c^* . Did leisure increase compared to LL-3?

Show your work below:

Problem 5: LL-5

Extra drill - leisure-loving preferences

Jordan: $T = 24$ hrs, $w = \$15/\text{hr}$, $V = \$60$.

$U(c, l) = c^{0.25} * l^{0.75}$ (more leisure-loving than Maria).

Use $MU_l / MU_c = w$ to find l^* , L^* , c^* .

Show your work below:

Problem 6: LL-6

Extra drill - wage comparative statics

Sam: $T = 10$ hrs, $V = \$100$, $U(c, l) = c^{0.5} * l^{0.5}$.

(a) If $w = \$10$, find l^* and L^* .

(b) If $w = \$40$, find l^* and L^* .

(c) As wage rises, does Sam work more or less?

Show your work below:

Answer Key

LL-1:

BC: $c = 20 \cdot (16 - l) + 80$. $l^* = 10$, $L^* = 6$, $c^* = \$200$

LL-2:

$l^* = 9.33$, $L^* = 6.67$, $c^* = \$360$. Works MORE (w up $\rightarrow$ l^* up but $L^* = T - l^*$ also up for $a = b = 0.5$)

LL-3:

$l^* = 4$, $L^* = 4$, $c^* = \$100$

LL-4:

$l^* = 4.8$, $L^* = 3.2$, $c^* = \$120$. Yes - more leisure

LL-5:

$l^* = 15.5$, $L^* = 8.5$, $c^* = \$187.50$

LL-6:

(a) $l^* = 6$, $L^* = 4$ (b) $l^* = 7.5$, $L^* = 2.5$ (c) Works LESS as w rises